



FAPS

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Fürst
Prof. Dr.-Ing. Florian Risch

Lehrstuhl für Fertigungsautomatisierung
und Produktionssystematik



Friedrich-Alexander-Universität
Technische Fakultät

Production processes for inductive charging systems

Vorlesung, Produktion elektrischer Motoren und Maschinen
Lecture, Production of electric motors and machines

Inductive charging systems are paving the way for a breakthrough in electromobility. Automated production in Germany has the potential to become a valuable new industry.

Inductive charging systems transfer electrical energy from one object to another without contact.

Energy is transmitted by electromagnetic fields without physical connection.

After attending the specialization unit “Production processes for inductive charging systems” students shall:

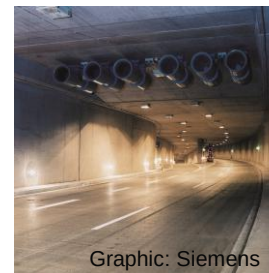
- **be familiar with the setup and construction of contactless energy transmission systems,**
- **understand functional principles of contactless power transmission systems for electric vehicles,**
- **know the manufacturing process chain for the production of the coil pairs used in an inductive power transmission system and understand the various process alternatives.**



Graphic: Omron



Graphic: ZF Friedrichshafen



Graphic: Siemens



Graphic: Kuka

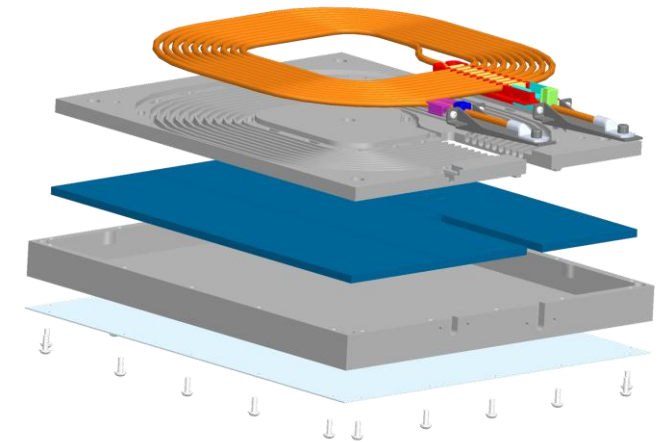
Inductive charging systems are paving the way for a breakthrough in electromobility. Automated production in Germany has the potential to become a valuable new industry.

■ **Fundamentals of inductive charging systems for electric vehicles**

- Motivation
- Types of energy transfer
- Design and functionality
- Challenges

■ **Production processes for the production of coil pairs**

- Semi-finished stock materials
- Coil winding process
- Electrical contacting
- Handling and assembly of ferrite bodies
- Potting



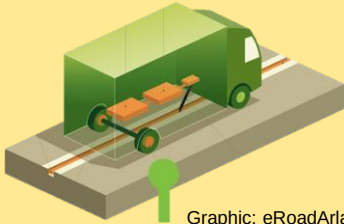
Wired charging solutions are not the ideal advertisement for electric vehicles as the mobility solution of the future in the public cityscape.



Problems associated with wired charging solutions

- Manual plug-in required before charging can begin
- Cable handling and stowing
- Tripping hazard in public spaces
- Vandalism (charging station, cables)
- Contamination or corrosion of the contact points

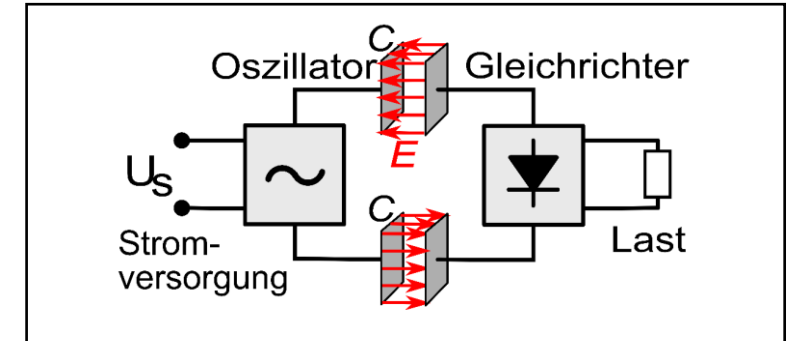
As an alternative to wired charging technologies, energy can also be transferred wirelessly to electric vehicles, including while driving.

	Stationary	Semi-dynamic	Dynamic	
Conductive			 Graphic: eRoadArlanda	 Graphic: Honda
Inductive				 Graphic: Traficverket
	At-home charging, Work, Shopping	Taxi stands, traffic lights, bus stops	Electrified lanes in highways	 Graphic: Siemens
				 Graphic: eRoadArlanda

Wireless energy transfer can be implemented on the basis of various operating principles.

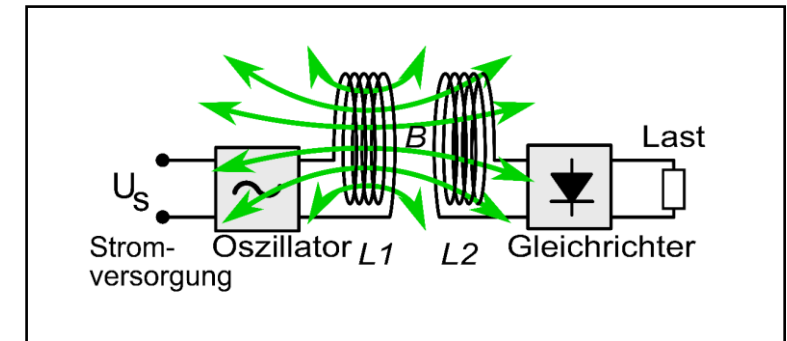
Capacitive coupling

- Energy transmission by alternating electric field.
- High power dissipation with larger plate spacing.



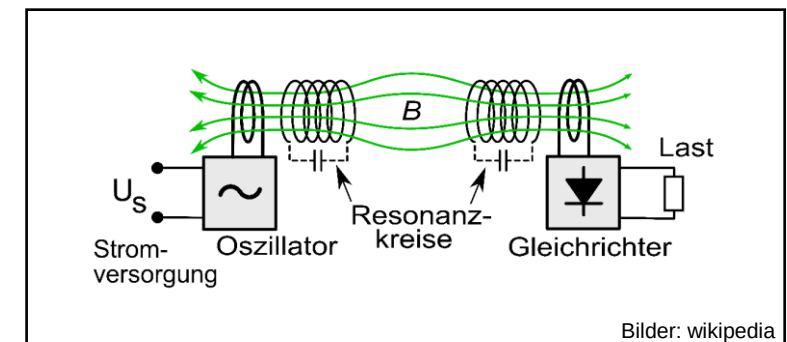
Inductive coupling

- Inverter generates high-frequency AC voltage.
- Transmitter coil L1 generates alternating magnetic field, which induces an alternating voltage in transmitter coil L2.
- AC voltage is rectified to supply secondary loads.



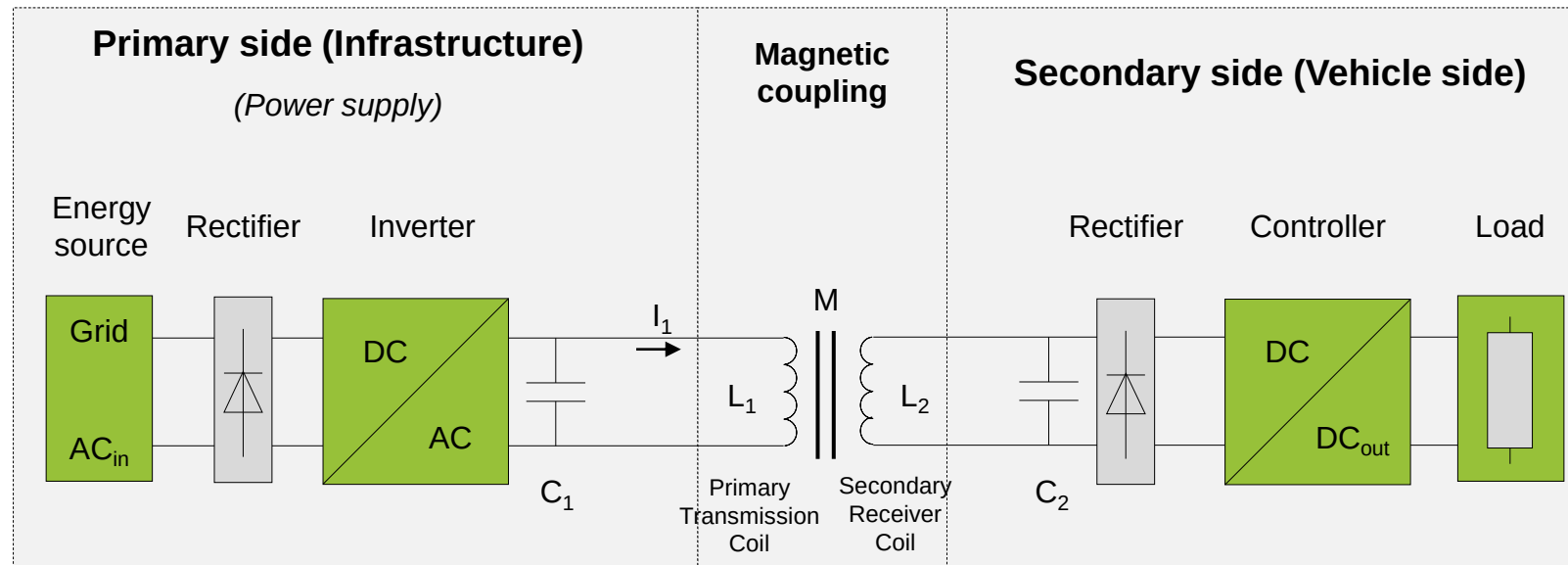
Resonant Inductive Coupling

- An oscillating circuit, consisting of capacitors and an inductive coil, is connected to both sender and receiver coils.
- Resonant frequency is matched to the transmission frequency.
- Resonance between oscillating circuits improves magnetic coupling of the coils and thus range and efficiency of transmission.



Bilder: wikipedia

An inductive charging system consists of both a primary and secondary side.
The topology can be represented in a block diagram.

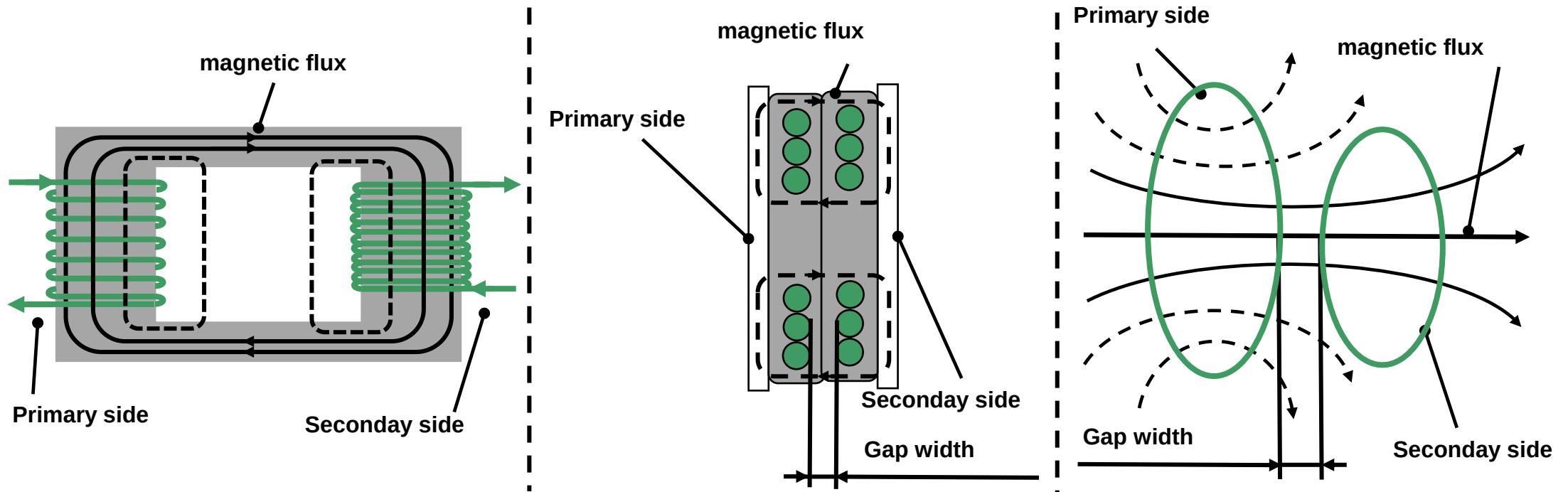


Topology of an IPT system

- Primary side components
 - Power supply (3-phase grid),
 - Rectifier,
 - Inverter
 - transmitter coil incl. oscillating circuit (compensation topology).
- Secondary side components
 - Receiving coil incl. oscillating circuit (compensation topology),
 - Rectifiers, DC Voltage Converters
 - Loads
- Magnetic coupling of primary and secondary coil

Depending on the application, the air gap between sender and receiver coils changes thus affecting the coupling factor of the coil system.

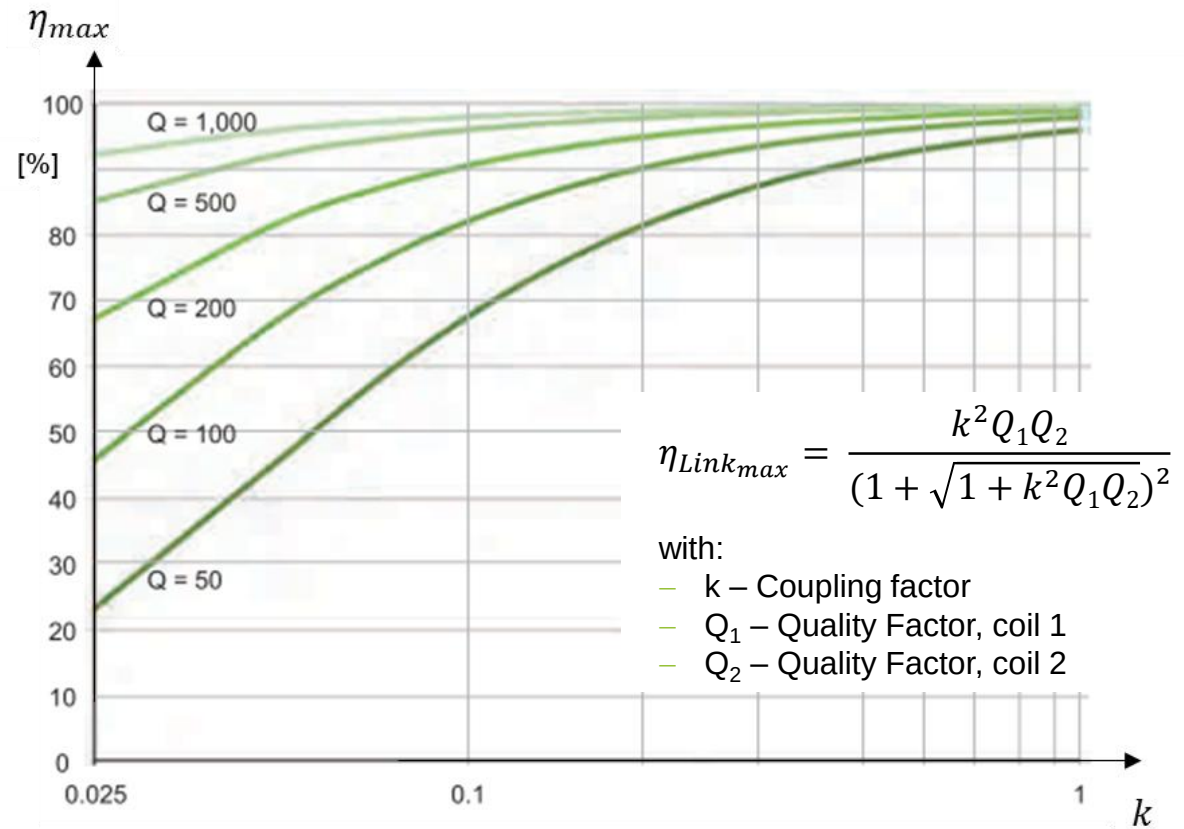
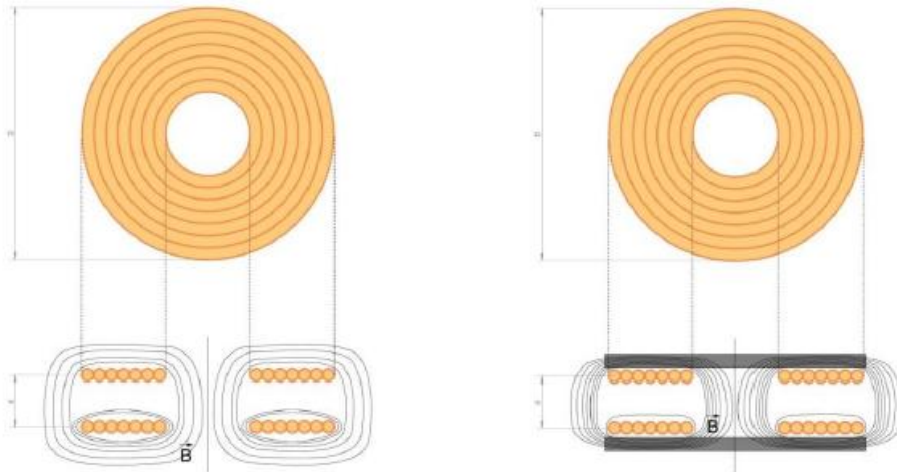
Coupling through closed iron core	Close coupling due to small air gap	Loose coupling due to large air gap
Coupling factor $k \sim 95\%$	Coupling factor $k \sim 50\%$	Coupling factor $k < 10\%$
Size of air gap = 0	Air gap width $\ll$ Coil diameter	Air gap width $\leq$ Coil diameter
Conventional transformers	Inductive charging pads, surface charging	Undercarriage system for EV charging



The system efficiency of a charging system can be estimated based on both the coupling factor and the quality factor of the coil pairs.

Influences on the coupling factor: k

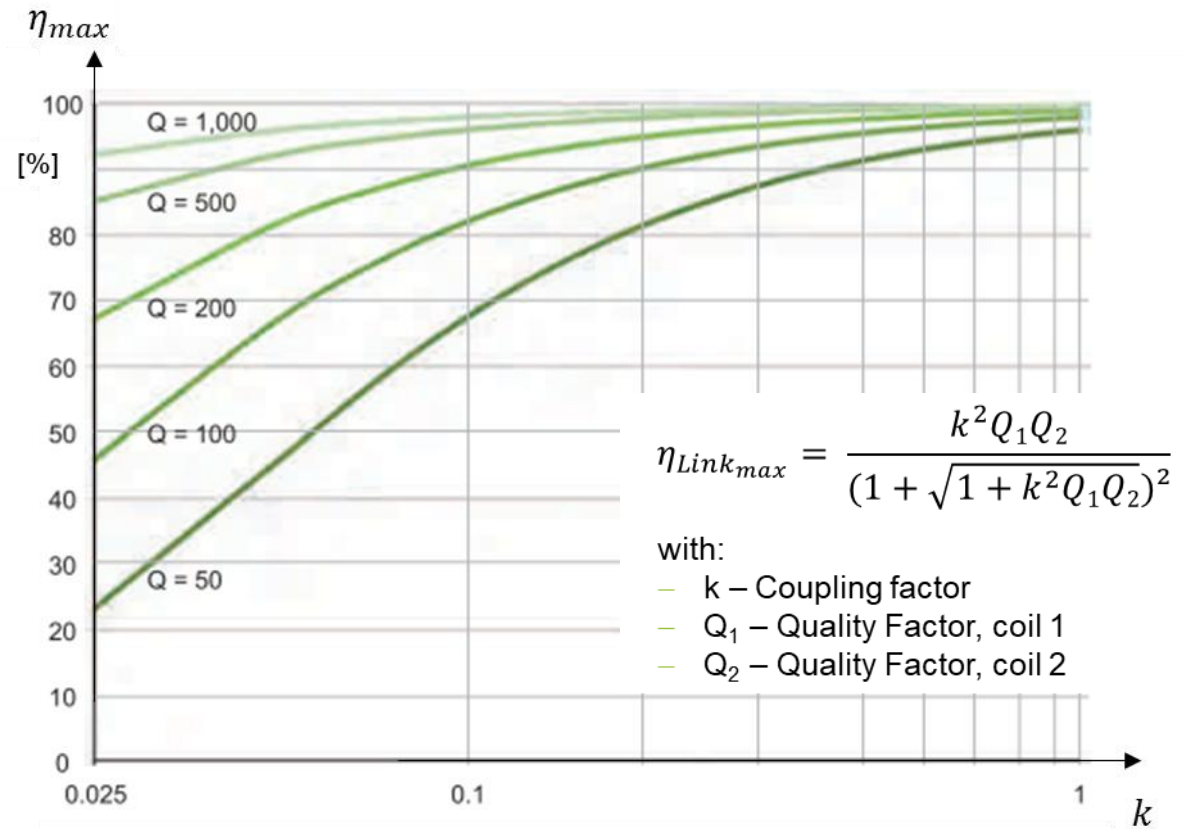
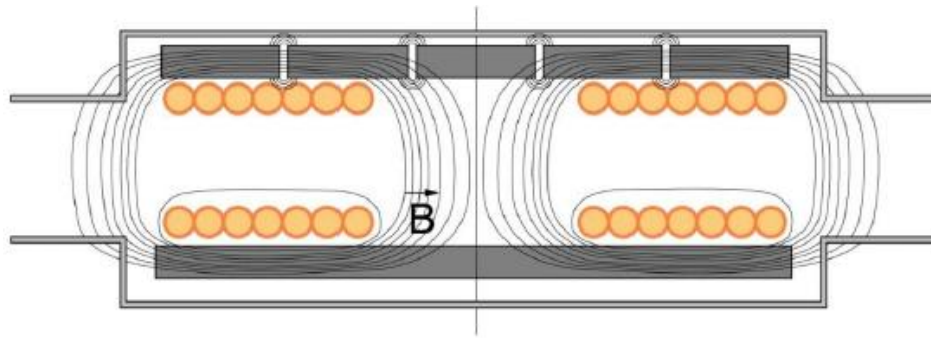
- Geometry (coil size, air gap) of the coil pair
- "Quality" of the magnetic field path



The system efficiency of a charging system can be estimated based on both the coupling factor and the quality factor of the coil pairs.

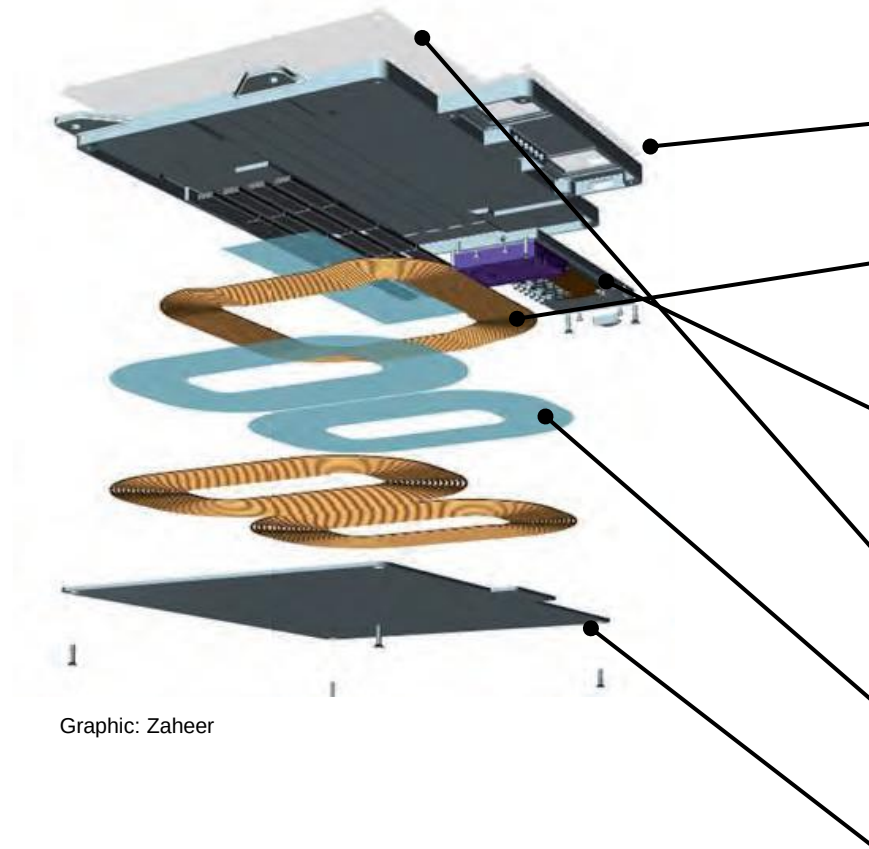
Influences on the quality factor: Q

- Magnetic field strength in the air gap (influenced by coil size and shape)
- Surroundings of the coil pair (ferrites for guiding the magnetic field, shielding)
- Electric resistance of the coil pair



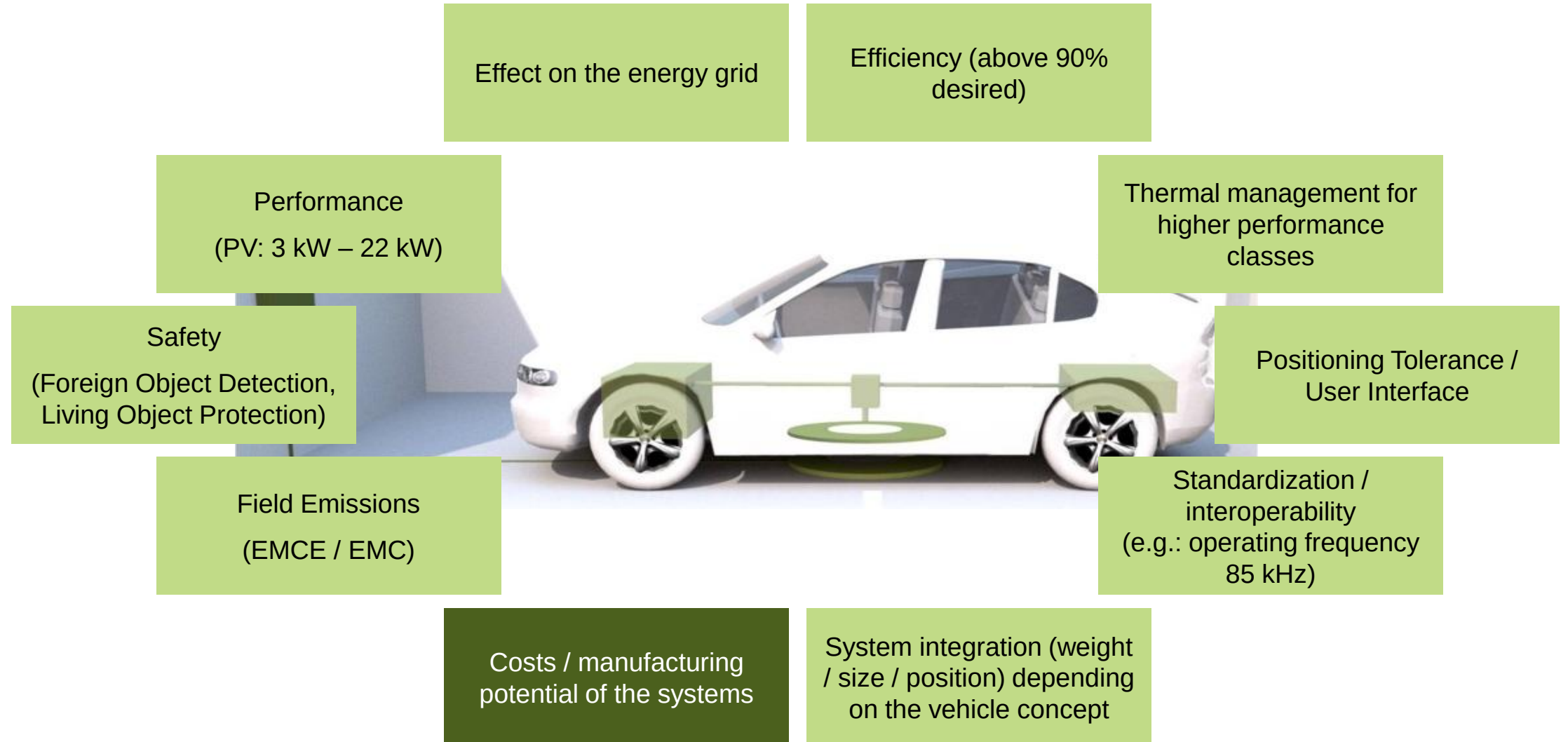
Inductive charging pads consist of various components necessary to fulfill a variety of functions.

Secondary side



Component	Main function
Housing	Mechanical protection
Coils	Generate magnetic field (primary) or generate induced voltage (secondary)
Power electronics	Generate / convert AC or DC current
Magnetic field guidance and shielding	Direct magnetic field towards paired coil and protect people/electronics
Secondary insulation	Electrical insulation and thermal coupling
Cover plate	Mechanical protection

Market adoption of contactless electric vehicle charging systems faces a number of complex challenges.



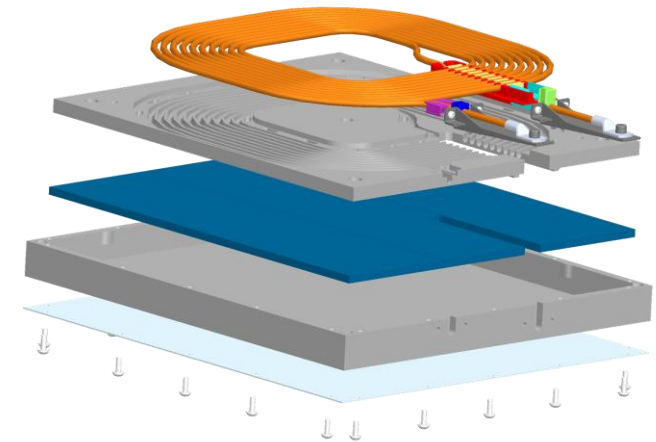
Inductive charging systems are paving the way for a breakthrough in electromobility. Automated production in Germany has the potential to become a valuable new industry.

■ **Fundamentals of inductive charging systems for electric vehicles**

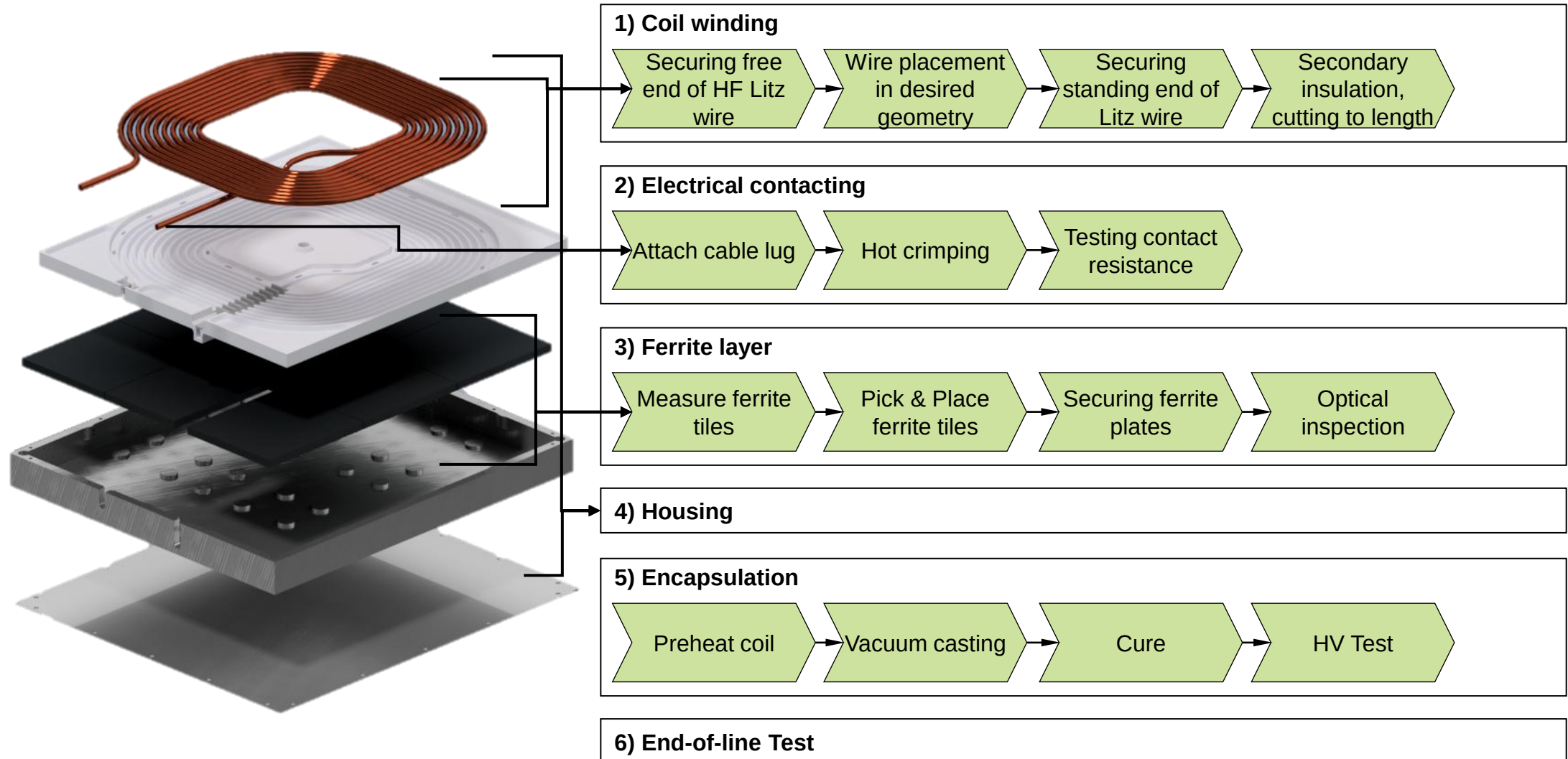
- Motivation
- Types of energy transfer
- Design and functionality
- Challenges

■ **Production processes for the production of coil pairs**

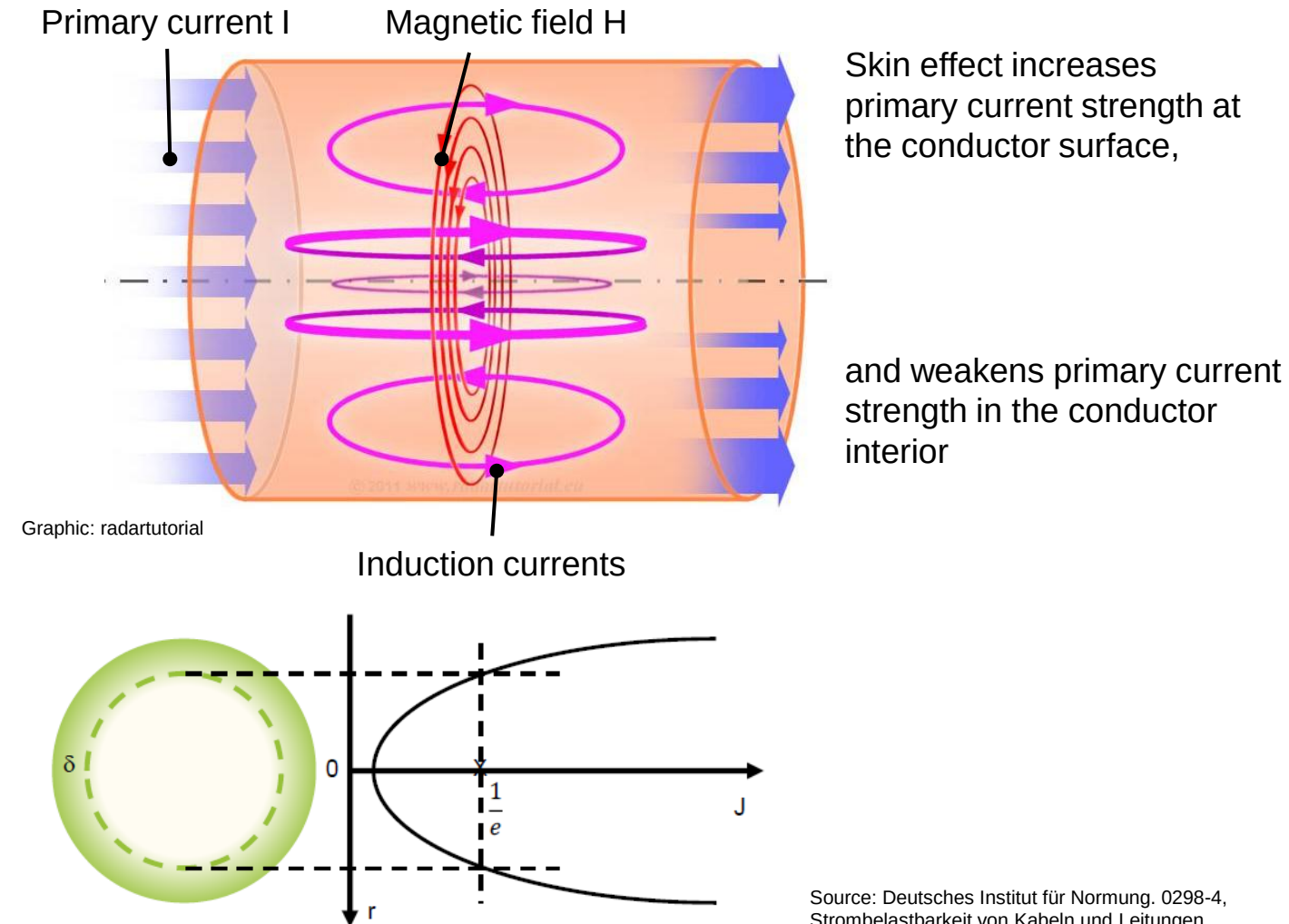
- Semi-finished stock materials
- Coil winding process
- Electrical contacting
- Handling and assembly of ferrite bodies
- Potting



The production process for the coil module of a contactless power transmission system consists of several individual processes.



Due to the occurrence of current displacement effects at high-frequency operation, stranded conductors are best suited for use at the operating frequency of 85 kHz.



Skin effect / Current displacement effect

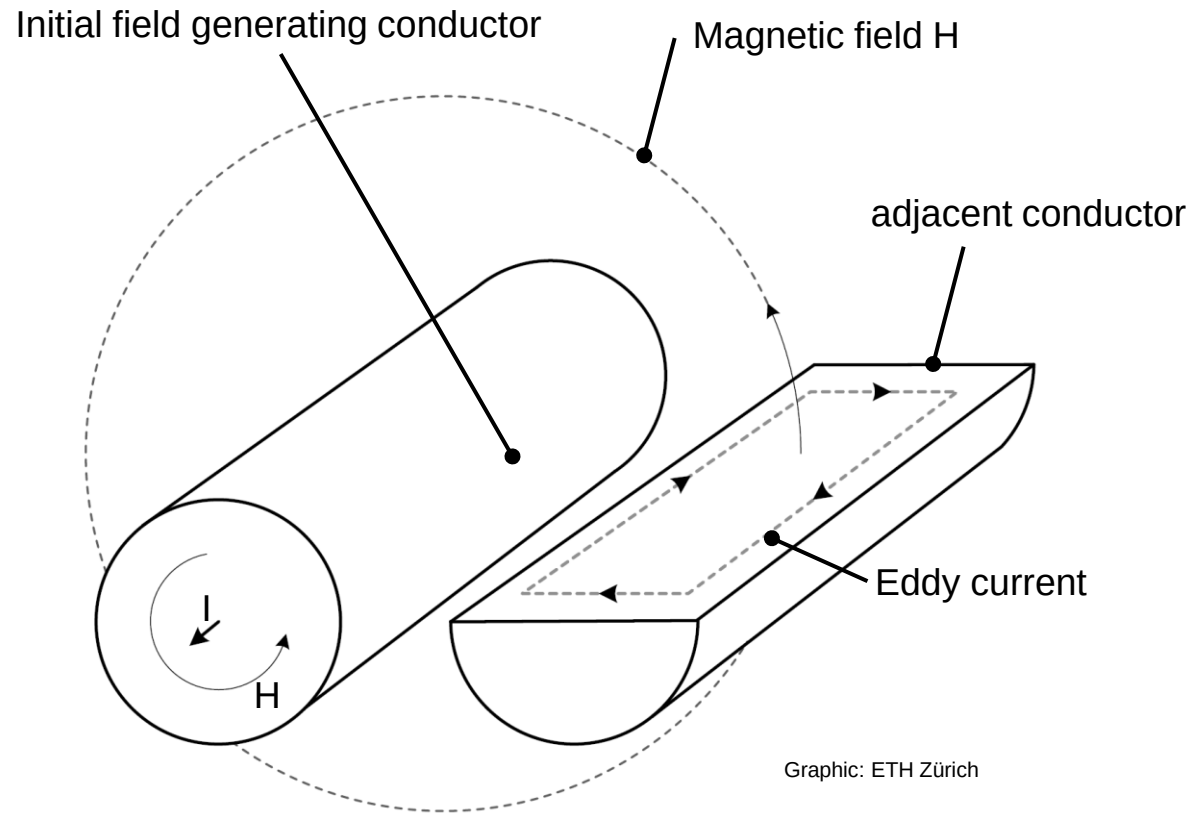
- Displacement of the current flow into the outer surface of the conductor
- Due to the flow of electricity, a magnetic field forms in the conductor, which in turn generates circular, superimposed eddy currents

- Approximate calculation of the Penetration depth from $\delta = \frac{1}{\sqrt{\pi\mu_0\sigma f}}$

with:

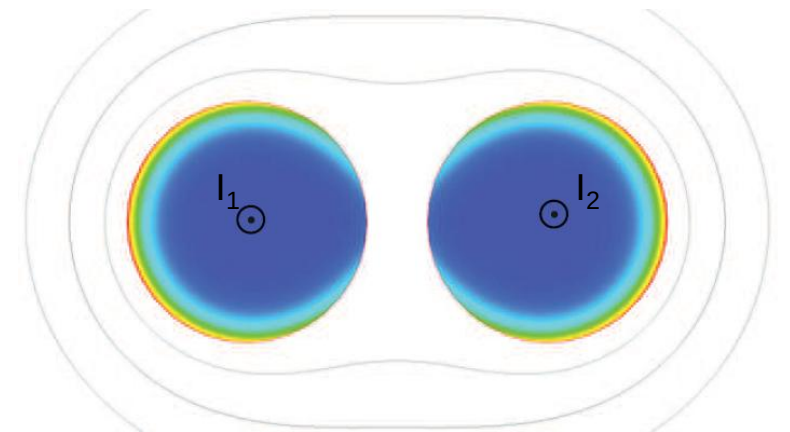
- μ_0 – Magnetic permeability constant
- $\mu_0 = 1,257 * 10^{-6} \frac{N}{A^2}$
- σ – Electrical conductivity of the material
- f – Frequency

The proximity effect describes current displacement due to the influence of adjacent conductors.



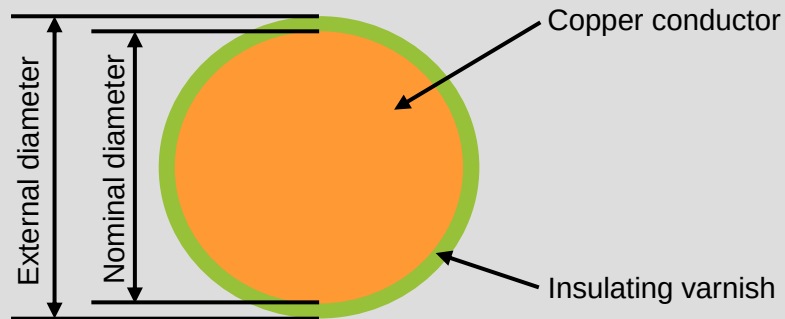
Proximity Effect

- Definition: The induced magnetic field created from a current-carrying conductor will generate eddy currents in adjacent conductors. The resulting eddy currents will cause current displacement in the conductor under consideration.
- External proximity effect: effect resulting from adjacent stranded conductors
- Internal proximity effect: effect resulting from adjacent strands within a single stranded conductor



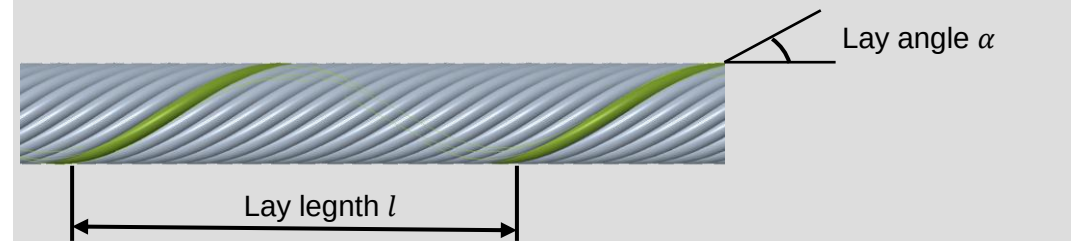
High-frequency Litz wire consists of many fine, individually insulated wires that are braided together to form larger strands.

Single strand made of copper conductor and insulating varnish

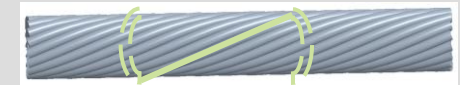


- Single strands are made of enameled copper wire according to standard series (DIN EN 60317)
- Single strand insulation: according to intended application often heat and solvent resistant varnish (ex. PAI, ...)
- Single strand diameter defined by strand designation. (ex.: 1050x0.1 corresponds to 1050 single strands, each with a diameter of 0.1 mm)

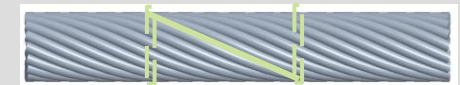
Basic terms for describing the geometry of an RF strand



Left-handed lay
Abbreviation: "S"



Right-handed lay
Abbreviation: "Z"

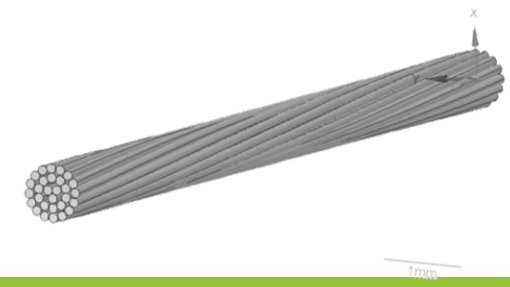


Bunched bundle	Concentric Litz	Semi-concentric Litz

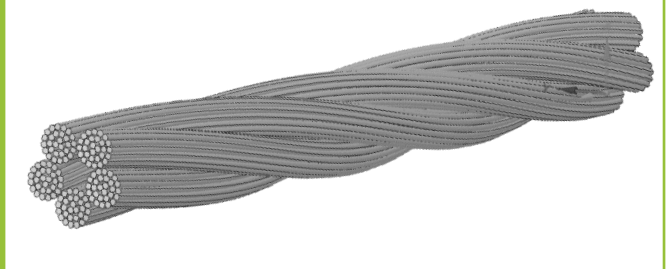
High-frequency Litz wire is defined by a variety of different parameters.

	Litz designation		Heermann 1740x0.1	1400x0.1	Leoni 1050x0.1	Leoni 4200x0.05
Strand	Strand material		-	Copper	Copper	Copper
	Single-strand diameter		mm	0.12	0.1	0.05
	Primary Insulation	Material	-	Polyurethane (PU)	Polyurethane (PU)	Polyamide imide (PAI)
		Isolationsgrad		GR2	GR1	GR1
Construction	Litz structure		-	semi-concentric	semi-concentric	semi-concentric
	Cabling step 1 (sub-bundel)	Number of strands	-	29	35	15
		Lay length	mm	32	20-30	20
		Lay direction	-	S (links)	S	Z
	Cabling step 2 (bundle)	Number of strands	-	5 x 29 = 145	10x35 = 350	5 x 15 = 75
		Lay length	mm	41	90	32,5
		Lay direction	-	S (links)	S	Z
	Cabling step 3 (Litze)	Number of strands	-	(4+8) x 5 x 29 = 1740	4x10x35 = 1400	(5+9) x 75 = 1050
		Lay length	mm	74.6	45	45
		Lay direction	-	S (links)	Z	S
Sheathing	Type		-	taped	None	wrapped
	Material		-	1xPET-Foil		Nylon
	Thickness		mm	0.023		0.05
	Overlap		%	50		50
Cross section		-	round	round	round	round
Conductor diameter		mm	5.8		5.3	4.9
Strand diameter (incl. sheathing)		mm	5.9			

Sub-bundle (29 strands)



Bundle (5 Sub-bundles)



HF-Litz (12 bundles)



The requirements for a high-frequency wire can be broken down into product and process requirements.

Perspective	Type of requirement	Requirement	Direction of optimization
Product / Operation	General	F Cost	▼
	Mechanical	F Size	▼
		F Fill factor	▲
		T Resistance to vibration (M-04 from LV 124*)	▲
		T Resistance to mechanical shock (M-05 from LV 124*)	▲
	Electromagnetic	F High-frequency resistance R_{DC} and R_{AC}	▼
		F Resistance to electrical breakdown	▲
	Thermal	F Thermal conductivity	▲
		F High-temperature resistance	▲
		F Resistance to heat shock (K-16 from LV 124*)	▲
	Chemical	T Resistance to salt spray test (K-06 from LV 124*)	▲

Perspective	Requirement	Direction of optimization for:		
		Winding	Contacting	Casting
Manufacturing process	F Tensile / compressive strength	▲		
	F Bending fatigue strength	▲		
	F Spring back	▼		
	F Insulation class		▼	
	F Ease of impregnation	▲		▲
	F Adhesion of secondary insulation	▲	▼	▲
	F High temperature resistance of conductor		▲	▲
	F Thermal decomposition of primary and secondary insulation		▲	

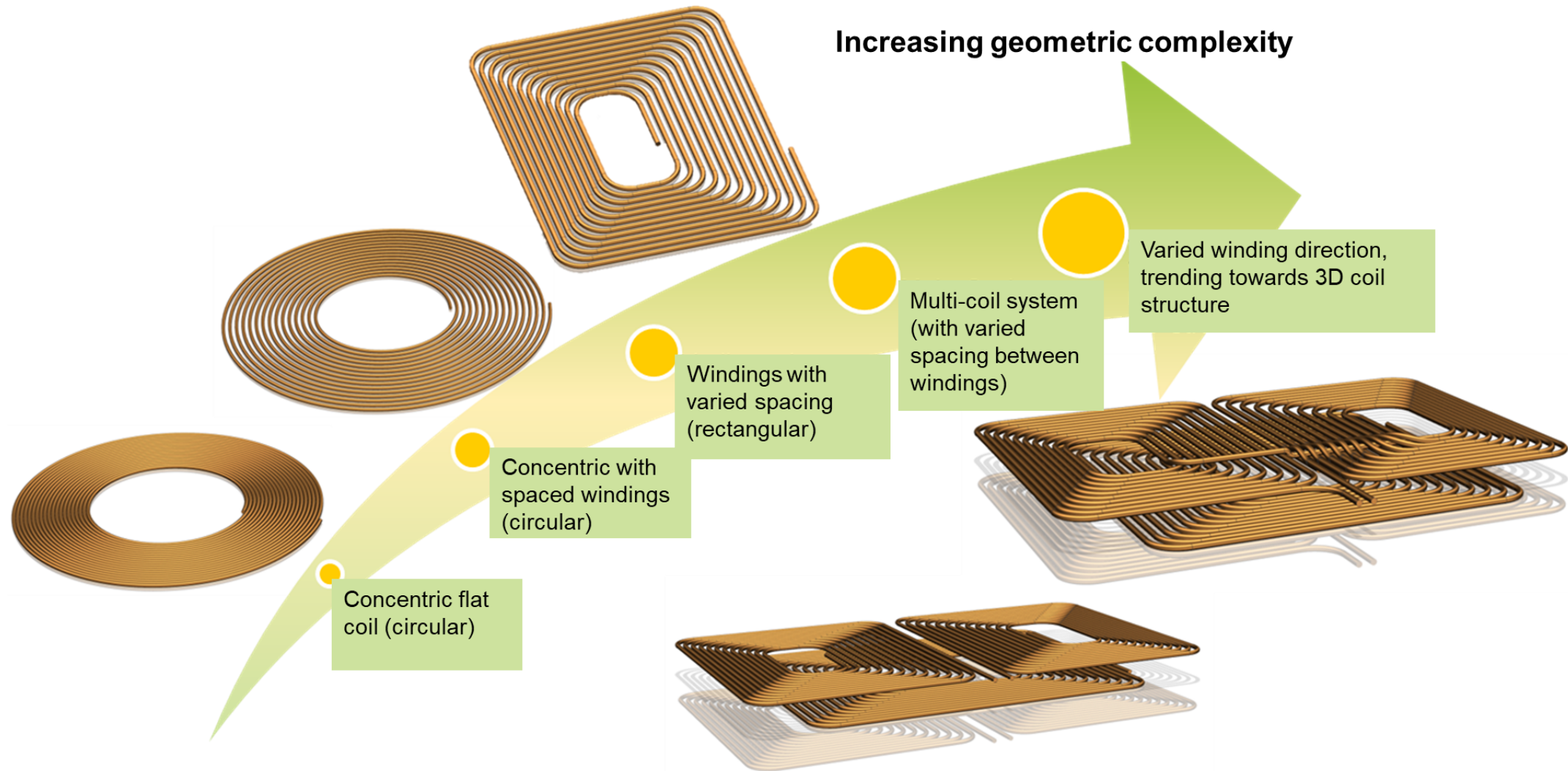
The requirements for a high-frequency Litz wire affect the configuration parameters.

QFD-Diagram (House of Quality):

- Interdependencies of requirements in the "roof" of the matrix
- Optimization directions of the quantifiable parameters of the Litz configuration displayed as arrows
- "+" corresponds to a positive effect while "-" corresponds to a negative effect
- Empty fields mean that there is no influence of the requirements on the Litz configuration

Optimierungsrichtung für		Produkt/Betrieb	▼	▼	▲	▲	▲	▼	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲	▲</
--------------------------	--	-----------------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

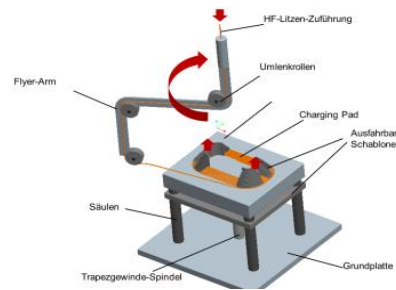
Flat coils made of HF-Litz wire can be manufactured using various processes dependant on coil geometry and complexity.



Flat coils made of HF-Litz wire can be manufactured using various processes dependant on coil geometry and complexity.

Concept I

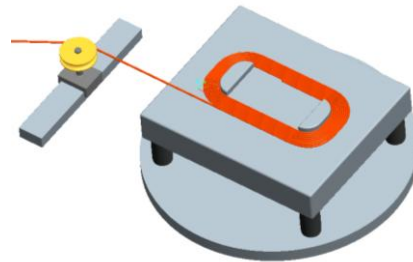
Flyer winding



Rotation of a flyer with a winding nozzle around a stationary fixed coil body

Concept II

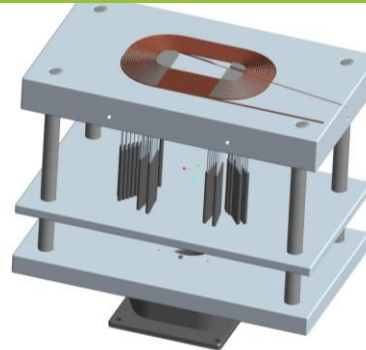
Linear winding



Winding with a winding nozzle onto a rotating coil form

Concept III

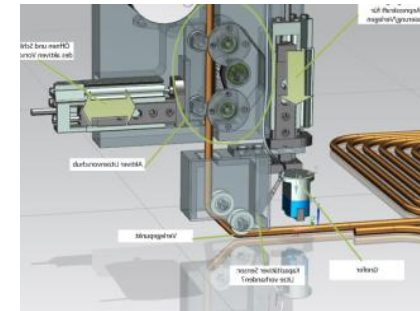
Linear winding with adaptive coil bodies



The shape of the coil is changed, e.g. by movable pins that emerge at different positions from a base.

Concept IV

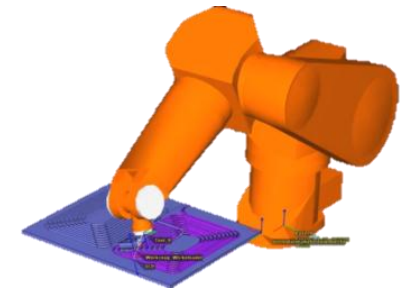
Placement onto base material



Laying and fastening the Litz wire onto a base material using special wire-laying devices

Concept V

Robot-based placement onto base material

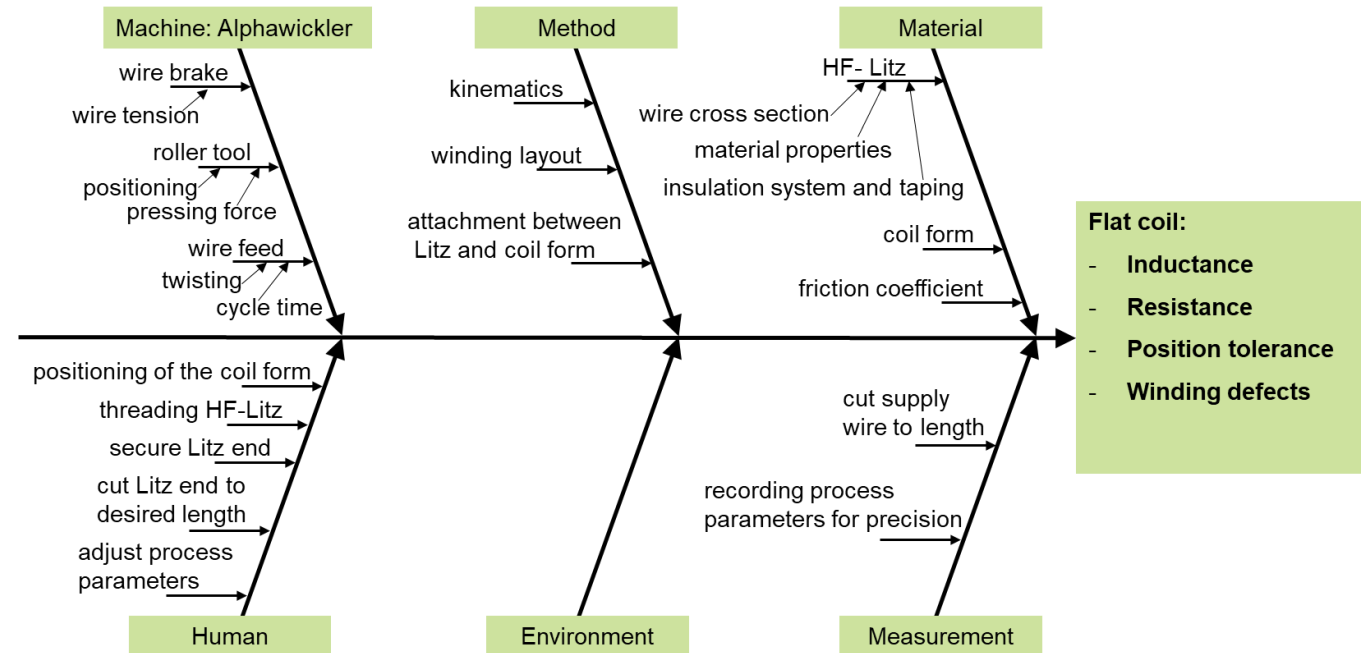


Robot-assisted laying and fastening of the Litz wire onto the base material.

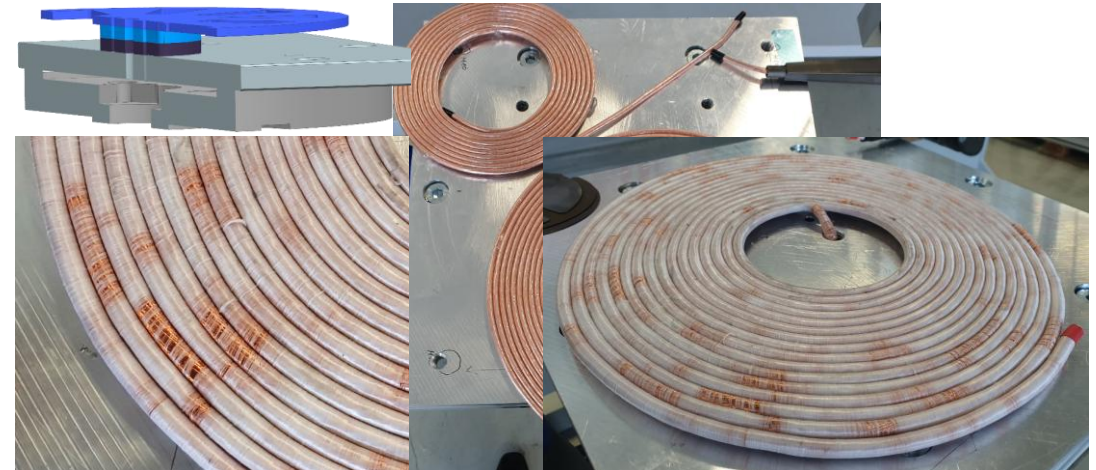
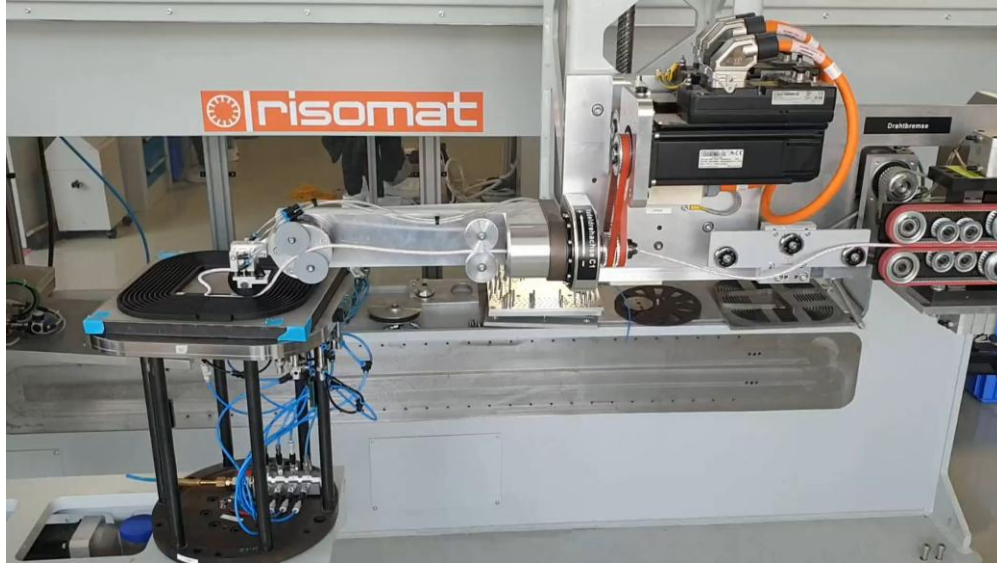
The winding process for flat coil production is subject to a variety of influencing parameters.

Winding process

- Flat coil production
- Implementation at FAPS: Universal winding machine, Risomat Alphawickler 2



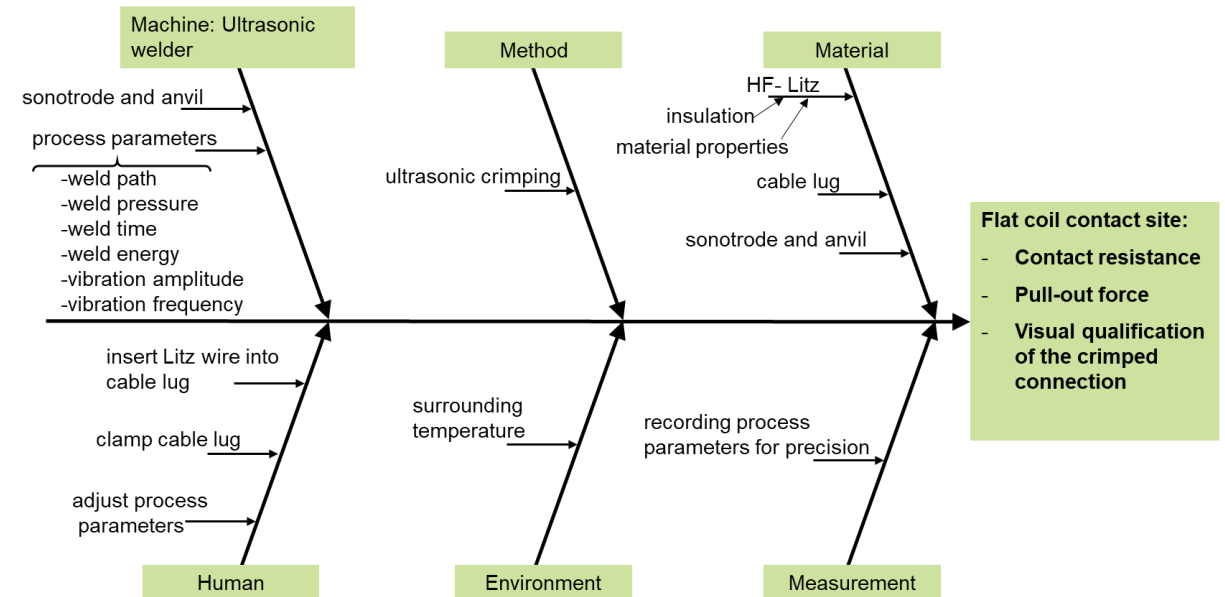
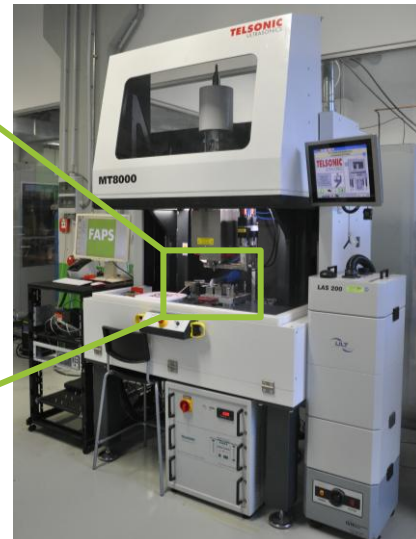
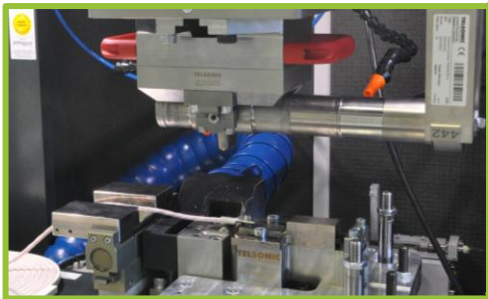
Flexible automated systems with many kinematic degrees of freedom enable the investigation and optimization of different coil arrangements and winding strategies.



The contacting process for creating electrical connections can be carried out using hot crimping or ultrasonic welding processes.

Contacting

- Creates contact points for the coil module
- Formation of a mechanical and electrical connection
- At FAPS: Ultrasonic welding machine MT8000 from Telsonic

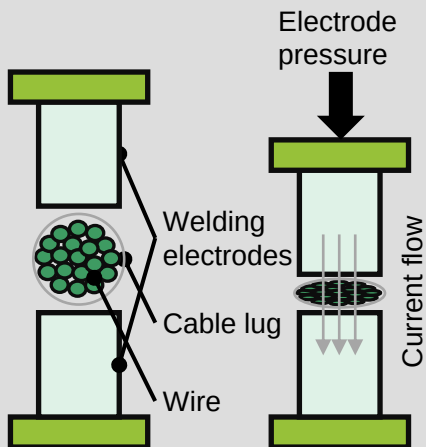


The hot crimping process reflects the current industrial standard for contacting enameled wire bundles.

Hot crimping

Hot crimping is a type of resistance welding in which a conductive connection part (a cable lug, terminal, or a simple round sleeve) is used.

- The generation of heat by the flowing current (Joule heating) leads to the vaporization of the insulation and at the same time to the cohesion of the wires
- By integrating the normal additional step of stripping, productivity in the production of electrical connections is increased



Advantages

- Stripping and contacting of enameled wire bundles
- Good automation and process control
- Good mechanical and electrical connection quality
- Short process times

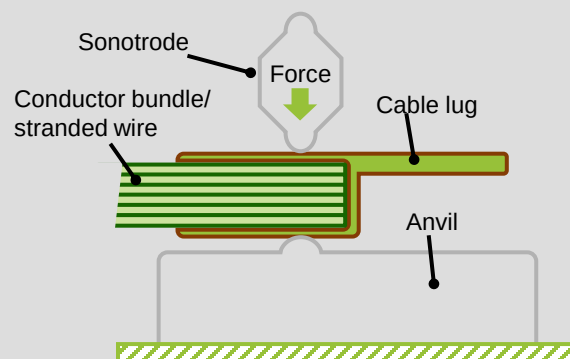
Disadvantages

- High tool wear (depending on electrode material)
- Combustion residues from the insulation
- High process energy requirement

Ultrasonic crimping is an energy-efficient alternative to the hot crimping process.

Ultrasonic crimping

- Combination of ultrasonic welding of metals with mechanical crimping
- Use of a connecting part (cable lug, sleeve, ...)
- Applied sonotrode force causes mechanical deformation of the connection part in the z-direction and cold welding of the joint
- High-frequency mechanical vibrations of the sonotrode in the range of 20 to 35 kHz simultaneously leads to friction between the connecting part and the enameled copper wires
- Generated thermal energy leads to melting and burning of the insulating layer



Video: FAPS



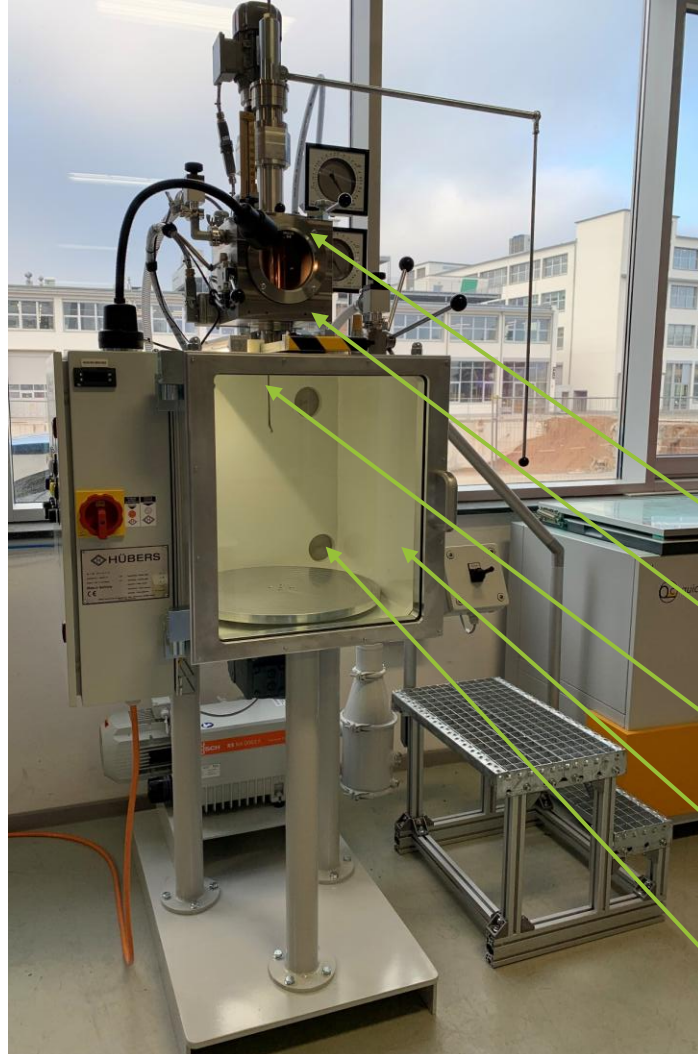
Advantages

- Stripping and contacting of enameled wire bundles
- Low process energy requirement
- Short process times
- Low tool wear

Disadvantages

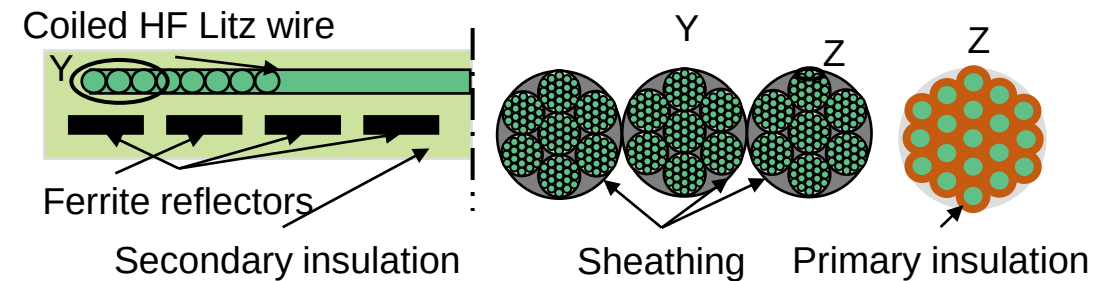
- Burning residues of the insulation
- Fatigue fractures of the cable lugs possible as a result of vibration coupling and damping effects

The durability of the coil module is largely determined by both the quality of the insulating material and the characteristic parameters of the potting process.

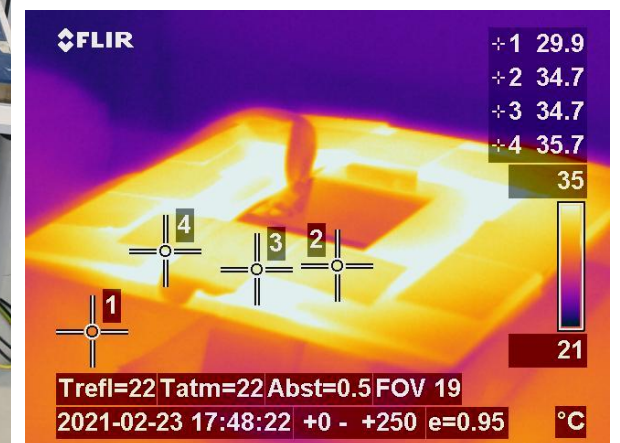
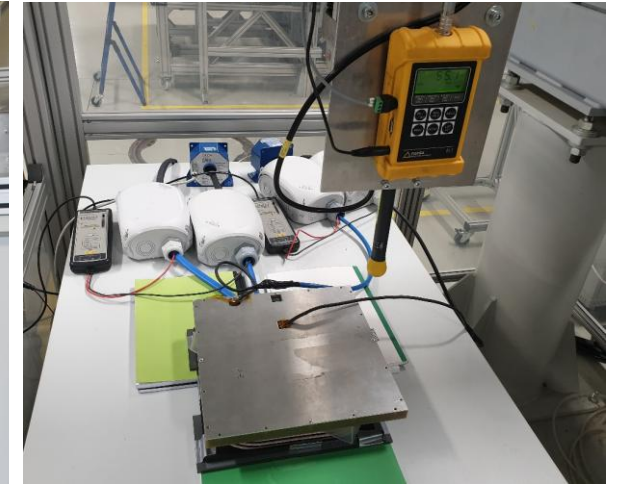
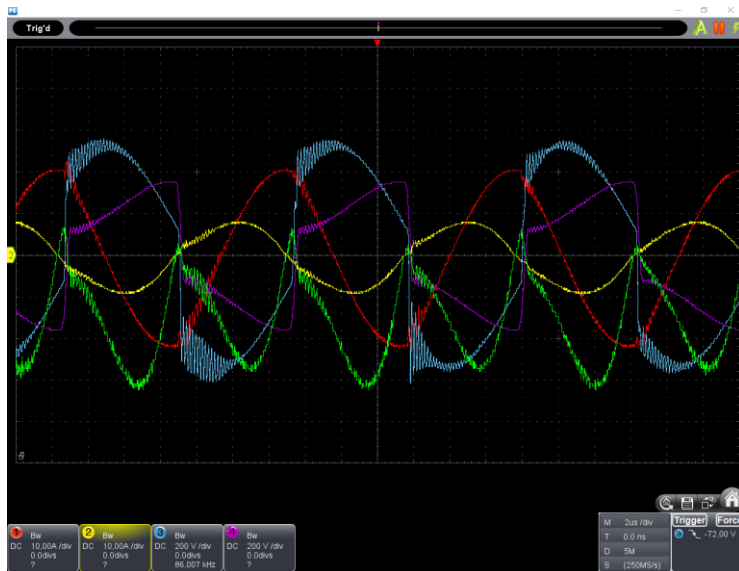


1. **Preparation** of the resin (Preheating, homogenizing, degassing, drying)
2. **Dosing** of resin and hardener according to prescribed weight ratio
3. **Mixing** of resin and hardener into a homogeneous potting compound
4. **Potting** of the coil and the ferrite cores with the epoxy potting compound
5. **Curing** of the resin by increasing the temperature

Primary and secondary isolation of IPT systems



- Primary insulation and sheathing ensures electrical properties
- Secondary insulation protects the coil structure from environmental influences (mechanical, thermal, chemical)



The research project E|ProFIL investigates efficient and automatable processes for the production of inductive charging systems.

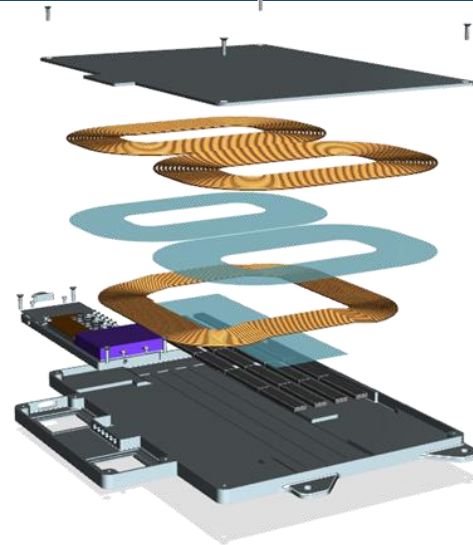
E|ProFIL: Efficient processes for the production of inductive charging systems

Product design

- Automation-compatible system design
- Maximization of system efficiency
- Minimization of space requirements

Semi-finished product design

- Processing optimized Litz wire design
- Processing optimized attachment concepts



Production process design

- Winding technologies
- Contacting technologies
- Insulation and fixation technologies
- Digitalization technologies

Gefördert durch:



Bundesministerium
für Wirtschaft
und Energie



Bundesministerium
für Umwelt, Naturschutz
und Reaktorsicherheit



DLR Projektträger

zollner



BOSCH

Technik fürs Leben

LEONI

FAPS

The manufacturing process chain of primary-side coil modules embedded into concrete for use in a electrified test road is the focus of the E|ROAD project.

Efficient manufacturing processes for an electrified road (E|ROAD)

Project goals and content

- Design of the coil modules suitable for production
- Automation-compatible processes for manufacturing the coil modules
- Integration of the coil modules into the roadway using precast concrete construction methods
- Construction of an electrified test track on the duraBAST demonstration site



[SELECT USU]



[Qualcomm]

Consortium

- **FAPS:** Manufacturing technologies of coil modules
- **Institute for Urban Engineering and Road Construction at TU Dresden:** Precast concrete components in road construction
- **wheelE GmbH:** Inductive energy transmission
- **BTE Stelcon GmbH:** Precast concrete components
- **Otto Alte Teigeler:** Jointing of precast concrete elements
- **Bast:** Mechanical load tests on demonstration site duraBAST

The contents of this lecture are discussed in further detail in the following resources:

Academic Papers

- A. Zaheer, G. A. Covic and D. Kacprzak, "A Bipolar Pad in a 10-kHz 300-W Distributed IPT System for AGV Applications," in IEEE Transactions on Industrial Electronics, vol. 61, no. 7, pp. 3288-3301, July 2014, doi: 10.1109/TIE.2013.2281167.
- A. Kuehl, M. Kneidl, J. Seefried, M. Masuch, M. Weigelt, and J. Franke, "Production concepts for inductive power transfer systems for vehicles," Energies, vol. 15, no. 21, p. 7911, 2022, doi: 10.3390/en15217911

Dissertations

- Risch, F. Planning and production concepts for contactless power transfer systems for electric vehicles. Zugl.: Erlangen-Nürnberg, Univ., Diss., 2014; Meisenbach: Bamberg, 2014, ISBN 9783875253696.
- Knaisch, K. Untersuchung von Spulensystemen zur induktiven Energieübertragung von Elektrofahrzeugen. Vergleich von Topologien und Entwicklung einer Auslegungsmethodik; KIT Scientific Publishing: Karlsruhe, Baden, 2017, ISBN 3731506300.





FAPS

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Fürst
Prof. Dr.-Ing. Florian Risch

**Institute for Factory Automation
and Production Systems**



Friedrich-Alexander-Universität
Technische Fakultät

DANKE